

Stability and Bifurcation of a Simple Food Chain in a Chemostat with Removal Rates

Numerical reproduction of El-Sheikh & Mahrouf (2005)

Vu Hai An Nguyen Dinh Truong Nguyen Nhat Minh
Le Minh Hai Doan Phuong Khang

Hanoi University of Science and Technology
School of Information and Communication Technology
Course: Math Modeling – Mentor: Dr. Bui Quoc Trung

Hanoi, June 2026



SOICT

Roadmap

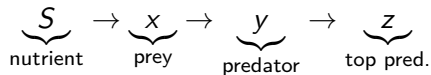
- 1 Model & Theory
- 2 Numerical Method
- 3 Theorem Validation
- 4 Robustness & Sensitivity

PART 1

Model & Theory

The problem

A chemostat food chain. Nutrient flows in and out at a fixed dilution rate; each level eats only the one below:



The paper (El-Sheikh & Mahrouf, 2005). One prey, **two** predators, each with a **distinct removal rate** D_i (washout + death).

Three questions: (1) who **survives**? (2) is coexistence **persistent**? (3) when does it start to **oscillate** (Hopf)?

Why it is hard

Distinct D_i break the conservation law $\Rightarrow$ the full **4-D system** must be analysed directly.

The model: four ODEs

State $X(t) = (S, x, y, z) \geq 0$; nutrient inflow and washout scaled to 1:

$\frac{dS}{dt} = 1 - S - f_1(S) x$	nutrient: inflow – washout – uptake
$\frac{dx}{dt} = x(f_1(S) - D_1) - f_2(x) y$	prey: growth – removal – eaten
$\frac{dy}{dt} = y(f_2(x) - D_2) - f_3(y) z$	predator
$\frac{dz}{dt} = z(f_3(y) - D_3)$	top predator

Each equation is a balance: gain from below – removal D_i – loss to the level above.

Monod (Michaelis–Menten) response for each link:

$$f_i(u) = \frac{a_i u}{k_i + u}, \quad f'_i(u) = \frac{a_i k_i}{(k_i + u)^2}, \quad f_i^{-1}(d) = \frac{d k_i}{a_i - d} \quad (d < a_i).$$

Assumptions on each f_i :

- $f_i(0) = 0$ — no food, no growth;
- $f'_i > 0$ — growth rises with food, then saturates.

Meaning. a_i is the max uptake rate, k_i the half-saturation ($f_i(k_i) = a_i/2$). The inverse f_i^{-1} gives the equilibria in closed form; a finite equilibrium needs $D_i < a_i$.

Equilibria: a trophic ladder

Four equilibria, one per regime:

Equilibrium	Survivors	Exists when
$P_0 = (1, 0, 0, 0)$	none	always
$P_1 = (S_1, x_1, 0, 0)$	x	$\lambda_x > 0$
$P_2 = (S_2, x_2, y_2, 0)$	x, y	P_1 exists & $\lambda_y > 0$
$P_3 = (S_3, x_3, y_3, z_3)$	x, y, z	P_2 exists & $\lambda_z > 0$

Invasion rates — can a missing species grow when rare?

$$\lambda_x = f_1(1) - D_1, \quad \lambda_y = f_2(x_1) - D_2, \quad \lambda_z = f_3(y_2) - D_3.$$

$\lambda > 0$: the next level invades; $\lambda < 0$: it washes out.

PART 2

Numerical Method

Implementation pipeline

Pure-Python, deterministic, built from first principles — every figure reproducible.

- ① **Read** the 9 parameters (a_i, k_i, D_i) for a scenario.
- ② **Integrate** the ODEs with RK4 (Euler kept as a baseline).
- ③ **Safeguard**: clamp tiny negative round-off to 0 and count it; flag explosions.
- ④ **Equilibria** as independent targets: Monod inverse for S_1, x_2, y_3 ; bisection on the monotone balance maps for the rest.
- ⑤ **Classify** the final regime ($P_0/P_1/P_2/P_3$) and compare with theory.

Each later tool (eigenvalues, elasticity) is introduced where it is first used.

Time integration: Euler and RK4

We solve $\frac{dX}{dt} = F(X; \theta)$ with two explicit fixed-step schemes ($h = \Delta t$):

Euler (1st order) — the baseline:

$$X_{n+1} = X_n + h F(X_n).$$

Runge–Kutta 4 (4th order) — used for all biological runs:

$$X_{n+1} = X_n + \frac{h}{6} (K_1 + 2K_2 + 2K_3 + K_4), \quad \begin{aligned} K_1 &= F(X_n), & K_2 &= F(X_n + \frac{h}{2} K_1), \\ K_3 &= F(X_n + \frac{h}{2} K_2), & K_4 &= F(X_n + h K_3). \end{aligned}$$

RK4 is far more accurate at the same step; Euler is kept to expose step-size error (Part 4).

Main simulation scenarios

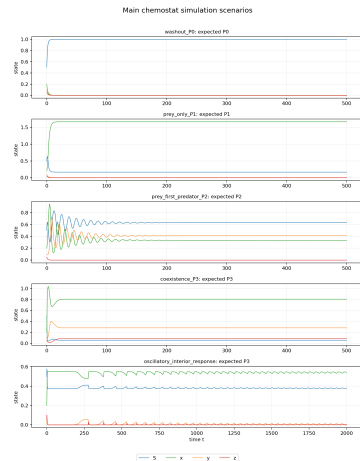
What we tested. We ran five representative long-time behaviours predicted by the equilibrium hierarchy.

Case	Interpretation
P_0	washout
P_1	prey-only survival
P_2	prey + first predator
P_3	steady coexistence
Oscillatory	coexistence near a Hopf boundary

How parameters were chosen. The parameter sets are fixed representatives, not random samples. We adjusted (a_i, k_i, D_i) so the signs of the invasion thresholds match the target regime:

$$f_1(1) - D_1, \quad f_2(x_1) - D_2, \quad f_3(y_2) - D_3.$$

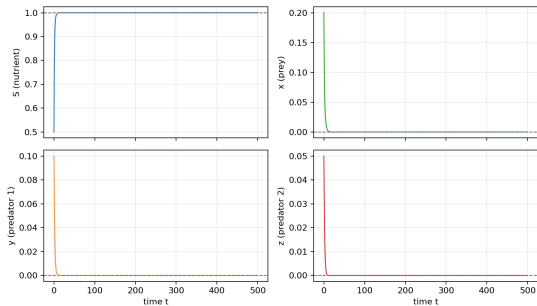
Positive sign means the next trophic level invades; negative sign means washout. The oscillatory case is chosen near the Hopf transition.



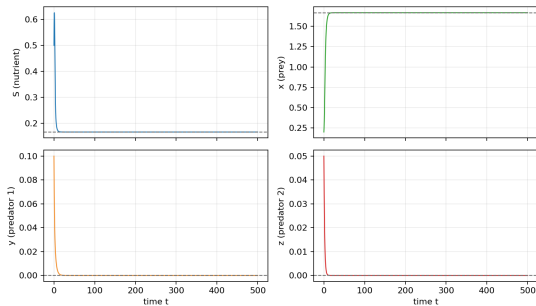
Simulation: washout and prey-only regimes

Each regime is integrated with RK4 from the same start; the final state reaches its analytic equilibrium.

washout_P0: Complete organism washout
(dashed line: theoretical equilibrium)



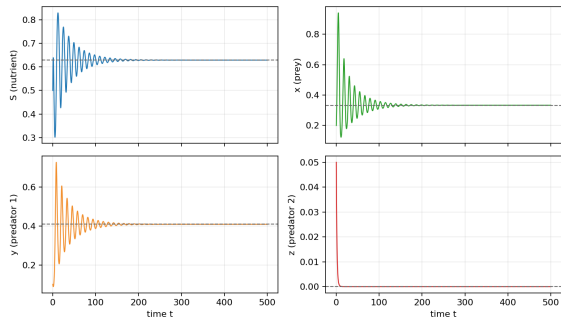
prey_only_P1: Prey survives; both predators wash out
(dashed line: theoretical equilibrium)



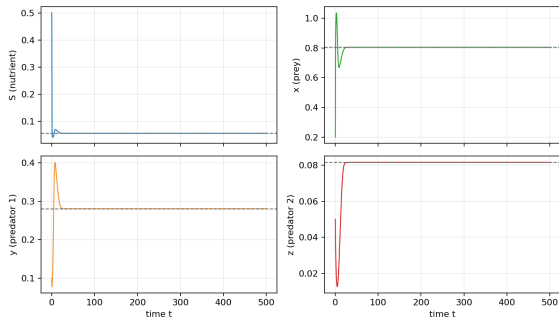
P_0 **washout** (left): prey cannot grow ($f_1(1) < D_1$), so all organisms vanish. P_1 **prey only** (right): prey survives, predators cannot invade ($\lambda_y < 0$).

Simulation: two-species and coexistence regimes

prey_first_predator_P2: Prey and first predator survive
(dashed line: theoretical equilibrium)



coexistence_P3: All trophic levels coexist
(dashed line: theoretical equilibrium)



P_2 two species (left): prey and first predator persist, top predator excluded ($\lambda_z < 0$). **P_3 coexistence** (right): all four levels settle at positive values.

Simulation: numerical vs. theoretical equilibria

RK4 final state compared with the analytic target (Euclidean distance):

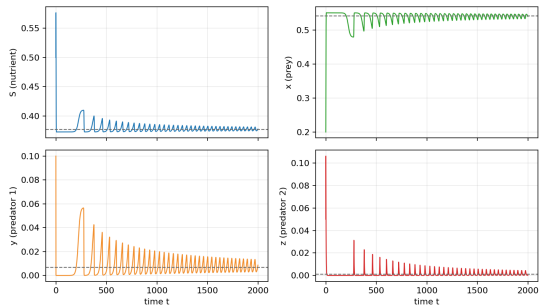
Scenario	Final state (S, x, y, z)	Distance
P_0 washout	(1, 0, 0, 0)	5.6×10^{-15}
P_1 prey only	(0.167, 1.667, 0, 0)	2.6×10^{-14}
P_2 two species	(0.629, 0.333, 0.409, 0)	1.3×10^{-6}
P_3 coexistence	(0.055, 0.805, 0.281, 0.082)	1.9×10^{-11}
Oscillatory (late mean)	(0.378, 0.540, 0.009, 0.000)	4.7×10^{-4}

Agreement to 6–15 significant figures — the simulator reproduces the equilibrium structure.

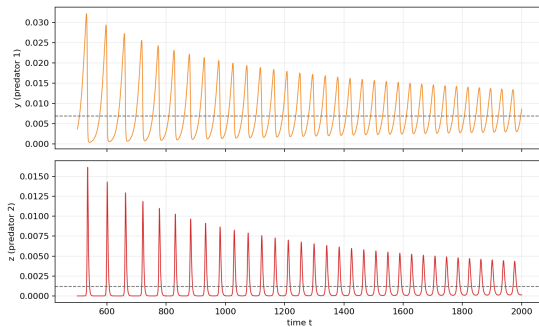
Simulation: slow oscillation near the Hopf boundary

The fifth scenario approaches its interior state through *slowly damped* oscillations — a preview of Experiment 4.

oscillatory_interior_response: Slowly damped oscillation around coexistence
(dashed line: theoretical equilibrium)



Oscillatory scenario: slowly decreasing predator-cycle amplitude



After $t = 500$: 23 peaks in y , 20 in z ; late/mid amplitude ratio ≈ 0.43 (still decaying); late-window mean 4.7×10^{-4} from P_3 .

PART 3

Theorem Validation

What the theory has to guarantee

So far we have only *watched* trajectories. Now we ask: does the maths *guarantee* that behaviour? Three questions, three experiments:

- ① **Boundedness.** Can any population blow up to infinity or run away? (No — everything is trapped in a finite region.)
- ② **Local stability.** If we nudge an equilibrium (a steady state where all four rates are zero, so nothing changes in time), does the system come back, or drift away?
- ③ **Hopf bifurcation.** As we slowly turn one knob, can a steady coexistence turn into a permanent oscillation?

The next slides give *just enough* theory to read the three result plots.

Theory — boundedness & dissipativity

Idea. Add the four ODEs: the predation terms cancel, leaving one equation for the *total mass* $T = S + x + y + z$:

$$\frac{dT}{dt} = 1 - (S + D_1x + D_2y + D_3z).$$

The removal term lies between $D_{\min}T$ and $D_{\max}T$ ($D_{\min}, D_{\max}$ = smallest / largest of D_1, D_2, D_3), so T grows when small and shrinks when large — and is eventually trapped:

$$T(t) \in \left[\frac{1}{D_{\max}}, \frac{1}{D_{\min}} \right] \text{ eventually.}$$

Every trajectory falls into one finite box and stays — the system is *dissipative* (nothing blows up). This is the sanity check behind every later experiment.

Theory — how we judge stability

Question. At an equilibrium, is the steady state stable (nudge it slightly and it returns) or *unstable* (nudge it and it drifts away)?

Linearise. Near an equilibrium a tiny perturbation δ grows like $\frac{d\delta}{dt} \approx J \delta$, where J is the *Jacobian* (the 4×4 matrix of slopes — how each of the four rates changes when each of the four variables changes):

$$J = \begin{pmatrix} -1 - f'_1 x & -f_1 & 0 & 0 \\ f'_1 x & f_1 - D_1 - f'_2 y & -f_2 & 0 \\ 0 & f'_2 y & f_2 - D_2 - f'_3 z & -f_3 \\ 0 & 0 & f'_3 z & f_3 - D_3 \end{pmatrix}.$$

stable $\iff \operatorname{Re} \lambda < 0$ for every eigenvalue λ
--

(every direction shrinks). If even one $\operatorname{Re} \lambda > 0$, that direction grows and the point is unstable.

Theory — the stability conditions

Two convenient facts. The Jacobian is *block-triangular* (the zeros in the corners), which splits the test into two easy pieces.

Piece 1 — invasion rate. For a species that is *absent* at the equilibrium, its own diagonal entry is its *invasion rate* (the growth rate of that species while it is still very rare):

$$\lambda_x = f_1(1) - D_1, \quad \lambda_y = f_2(x_1) - D_2, \quad \lambda_z = f_3(y_2) - D_3.$$

If $\lambda > 0$ the missing predator can invade $\Rightarrow$ the point is a *saddle*. This is exactly what pushes the chain *up* the trophic ladder.

Piece 2 — surviving block. For the species that *are* present we still need all their eigenvalues in the left half-plane. Rather than compute roots, we use *Routh–Hurwitz* (simple tests on the polynomial coefficients that are true *exactly when* every root has $\text{Re } \lambda < 0$):

$$P_0 : \lambda_x < 0; \quad P_2 : a_1, a_3 > 0, \quad a_1 a_2 > a_3 \text{ and } \lambda_z < 0; \quad P_3 : d_1, d_3, d_4 > 0, \quad \Phi > 0.$$

Theory — the Hopf quantity Φ

Where d_1, d_2, d_3, d_4 come from. They are the coefficients of the *characteristic polynomial* of the Jacobian — a degree-4 polynomial

$$\lambda^4 + d_1\lambda^3 + d_2\lambda^2 + d_3\lambda + d_4 = 0,$$

whose *roots are exactly the eigenvalues*; so $d_1, \dots, d_4$ are just the matrix repackaged as a polynomial.

The stability margin. For a degree-4 system the decisive Routh–Hurwitz number is

$$\Phi = d_1 d_2 d_3 - d_3^2 - d_1^2 d_4.$$

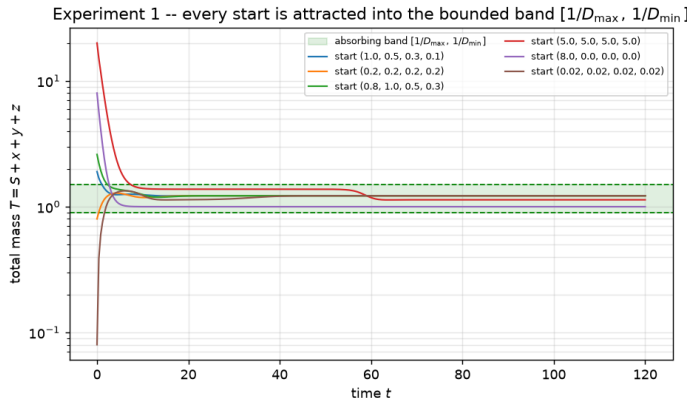
While $\Phi > 0$ the coexistence state P_3 is stable.

Hopf bifurcation. As we turn μ (here the prey uptake a_1), Φ decreases and hits 0 at μ^* . At that instant a *pair* of complex eigenvalues crosses the imaginary axis (eigen value = 0) *transversally* (at nonzero speed). The steady point loses stability and is replaced by a *limit cycle* (a closed loop the system circles forever)

Experiment 1 — Boundedness

Test. integrate 6 starts spanning very different total masses — including far *above* ($T_0 = 20$) and far *below* ($T_0 = 0.08$) the band $[0.90, 1.50]$.

- every start is drawn into the band and stays;
- no component ever becomes negative.



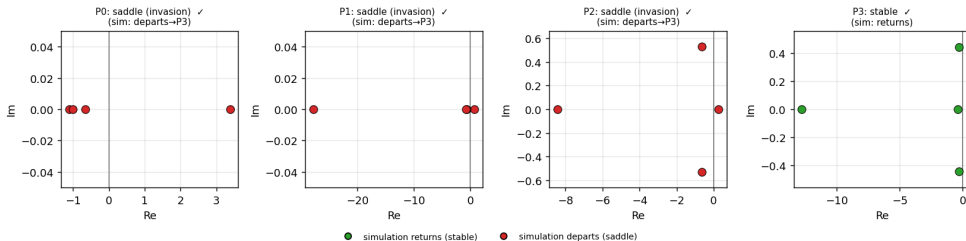
Experiment 2 — Local stability

Test. three independent verdicts at every equilibrium:

- analytic invasion-rate / Routh–Hurwitz conditions;
- sign of the largest Jacobian eigenvalue
- direct simulation nudged off the point.

All 36 equilibria agree(with 13 scenarios); boundary equilibria are saddles toward the next predator.

Experiment 2 -- local stability (coexistence_P3): dots = eigenvalues, colour = simulation, title = theory (✓ = all three agree)



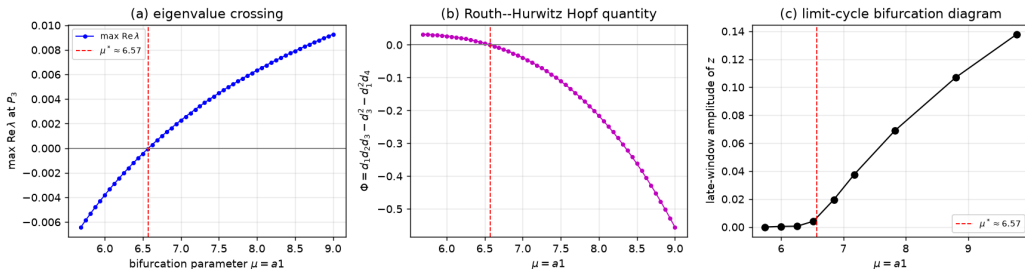
Green: all eigenvalues stable • Red: a right-half-plane eigenvalue (saddle).

Experiment 3 — Hopf bifurcation

Test. sweep $\mu = a_1$ (prey uptake) through the oscillatory scenario:

- **linear:** at $\mu^* \approx 6.57$ a complex pair crosses the axis — $\max \operatorname{Re} \lambda$ and Φ change sign together, transversally;
- **nonlinear:** below μ^* it settles to P_3 ; above, a *limit cycle* grows continuously from zero.

Experiment 4 -- Hopf bifurcation of the coexistence equilibrium P_3



Result: stable coexistence $\rightarrow$ periodic coexistence (supercritical Hopf).

PART 4

Robustness & Sensitivity

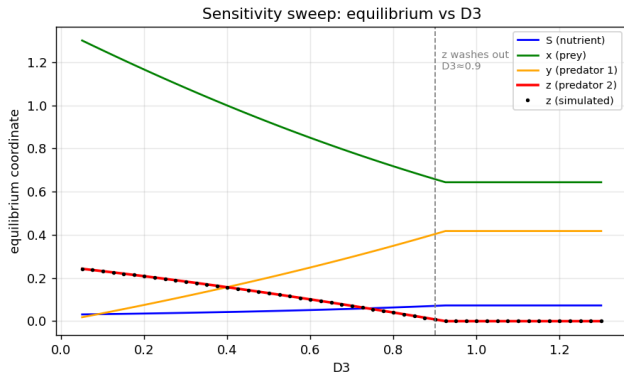
Removal rates decide the regime

Vary one removal rate, hold the rest.

Theory = simulation at all 138 points.

- $D_3 \uparrow$: top biomass z_3 falls *linearly* to 0 at $D_3 \approx 0.9$ ($P_3 \rightarrow P_2$);
- $D_1 \uparrow$: full cascade
 $P_3 \rightarrow P_2 \rightarrow P_1 \rightarrow P_0$; prey gone at $D_1 \approx f_1(1) \approx 4.1$.

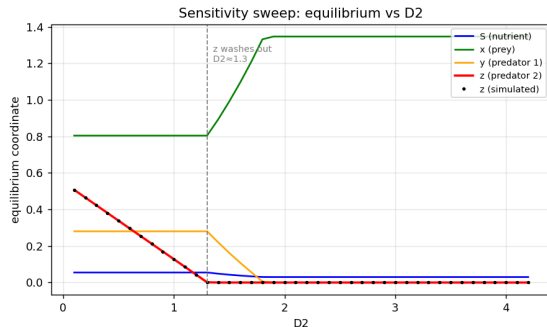
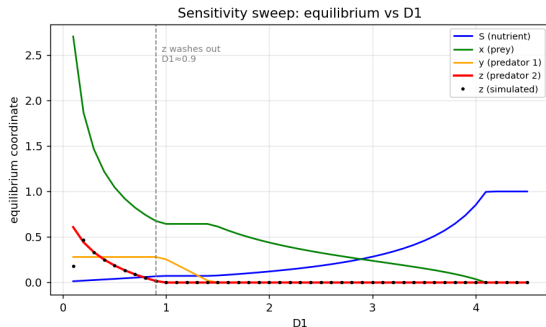
Increasing any rate removes levels from the top down.



Equilibrium coordinates vs. the top-predator rate D_3 .

Prey and first-predator removal rates

Same one-at-a-time sweep for D_1 (prey) and D_2 (first predator):



D_1 (left): drives the whole cascade $P_3 \rightarrow P_2 \rightarrow P_1 \rightarrow P_0$; prey persists until $D_1 \approx f_1(1) \approx 4.1$. D_2 (right): removes the two predator levels ($P_3 \rightarrow P_2 \rightarrow P_1$).

The top predator is the most fragile

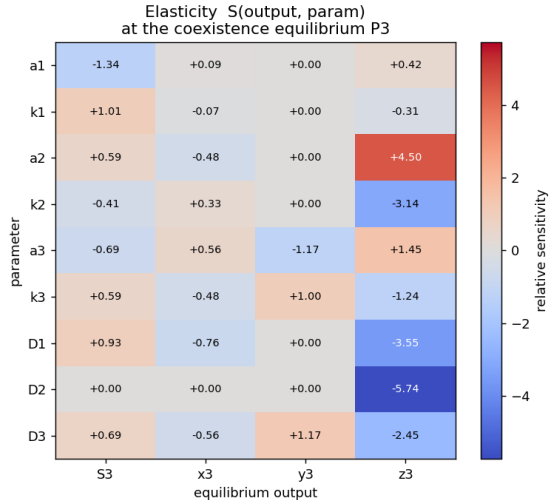
Elasticity

$$S(q, p) = \frac{p}{q} \frac{\partial q}{\partial p}$$

central difference (1% perturbation) — %
output change per 1% parameter change.

- z_3 is by far the most sensitive:
 $S(z_3, D_2) = -5.74$, $S(z_3, a_2) = +4.50$;
- y_3 reacts *only* to f_3 — a structural check,
since $y_3 = \frac{D_3 k_3}{a_3 - D_3}$.

The top of the chain accumulates every link
below $\Rightarrow$ first to be lost.



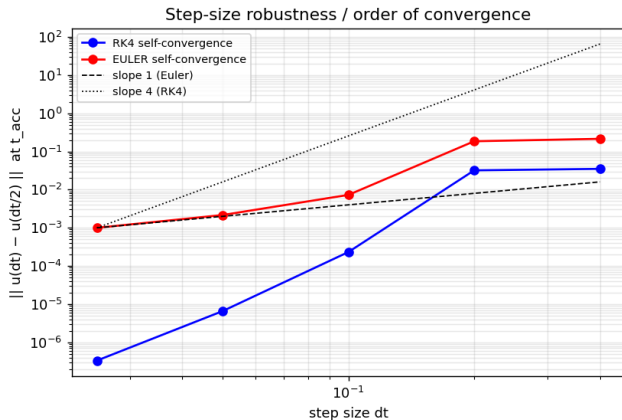
Signed elasticity (red +, blue -).

Robustness: RK4 vs. Euler

Step-halving convergence for both solvers:

- empirical order ≈ 4 (RK4), ≈ 1 (Euler);
- RK4: correct P_3 at every step, 0 clamps;
- coarse Euler: many clamps $\rightarrow$ **wrong** regime P_0 (spurious washout).

A poor integrator predicts extinction where the true dynamics coexist — the clamp count flags it automatically.



Self-convergence under step halving.

Step size in detail: RK4 succeeds, Euler fails

convergence diff at a short horizon; regime and clamp count at a long horizon.

Method	Δt	diff	regime	clamps	physical?
RK4	0.400	3.5×10^{-2}	P_3	0	SUCCESS
RK4	0.200	3.2×10^{-2}	P_3	0	SUCCESS
RK4	0.100	2.3×10^{-4}	P_3	0	SUCCESS
RK4	0.050	6.6×10^{-6}	P_3	0	SUCCESS
RK4	0.025	3.3×10^{-7}	P_3	0	SUCCESS
Euler	0.400	2.1×10^{-1}	P_0	16	FAIL
Euler	0.200	1.9×10^{-1}	P_0	83	FAIL
Euler	0.100	7.3×10^{-3}	P_3	1	FAIL
Euler	0.050	2.2×10^{-3}	P_3	0	SUCCESS
Euler	0.025	1.0×10^{-3}	P_3	0	SUCCESS

Coarse Euler reaches the wrong regime P_0 (16, 83 clamps); RK4 reaches P_3 at every step with 0 clamps.

Thank you!



SOICT